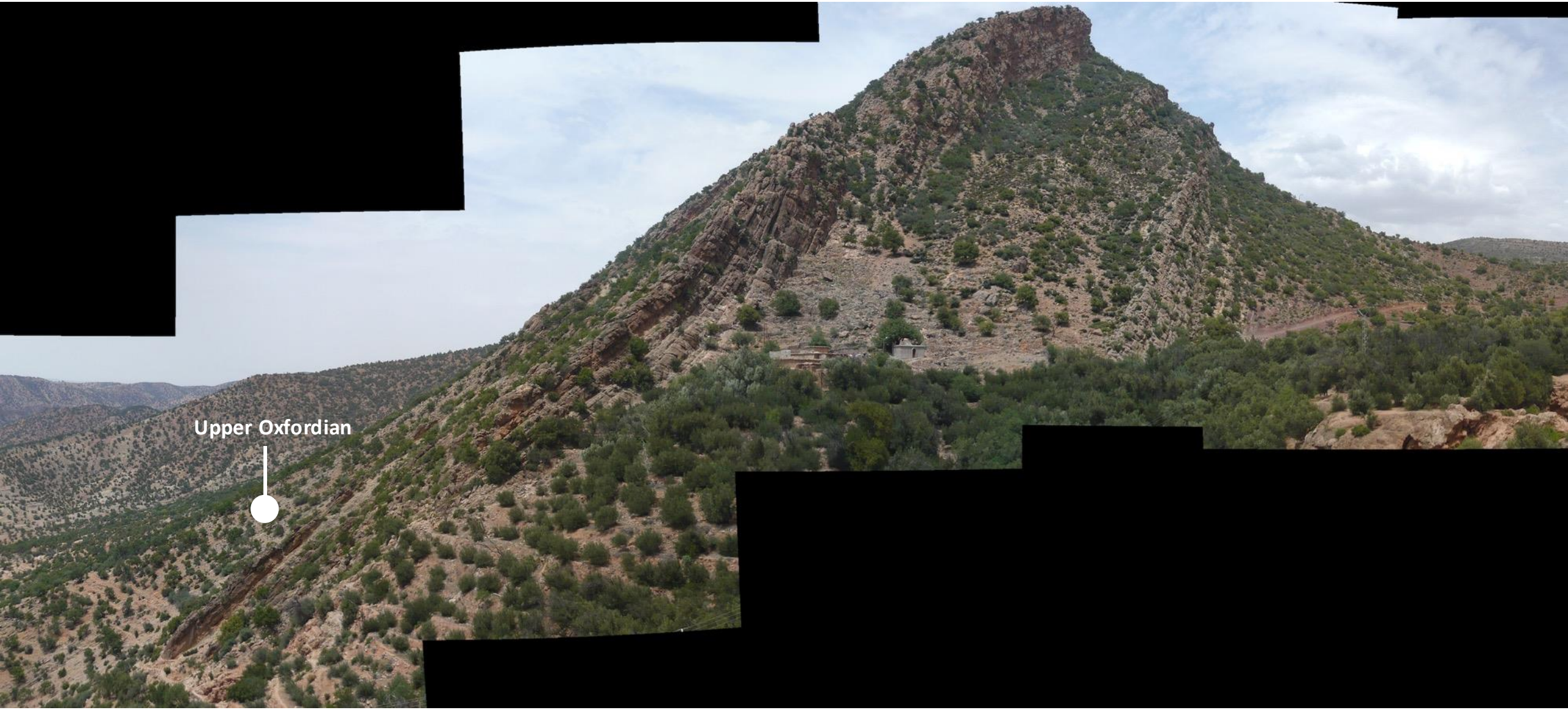
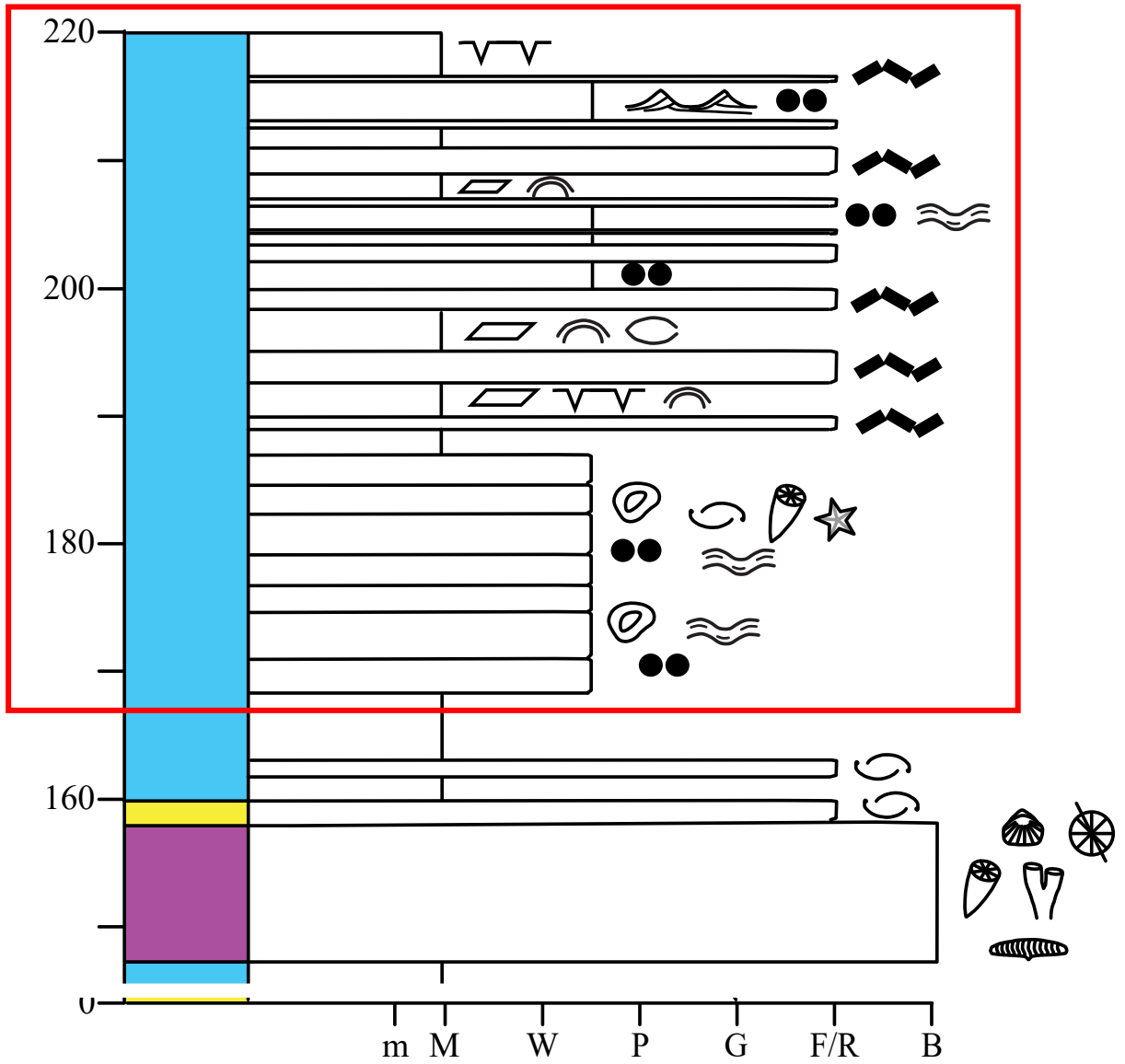
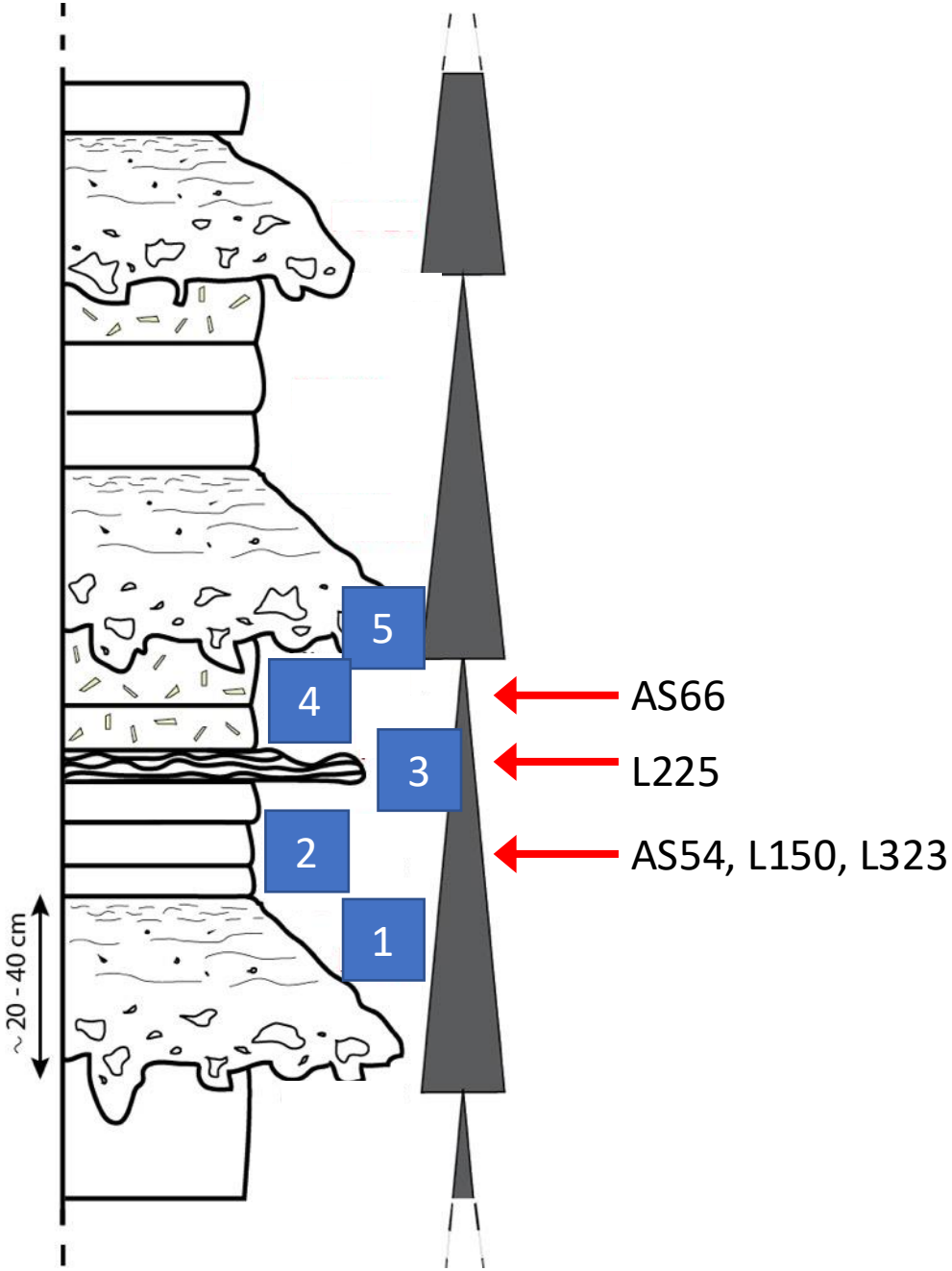
Outcrop overview



Study the following data set of utcrop photos and thin sections

Summarize your observations on this unit:
- What grains were identified?
- What sedimentary structures were identified?
- What is the texture of these beds?
- What sedimentary processes can you interpret?
- What environmental information is contained within the skeletal grains?
- Overall, what environment is likely?

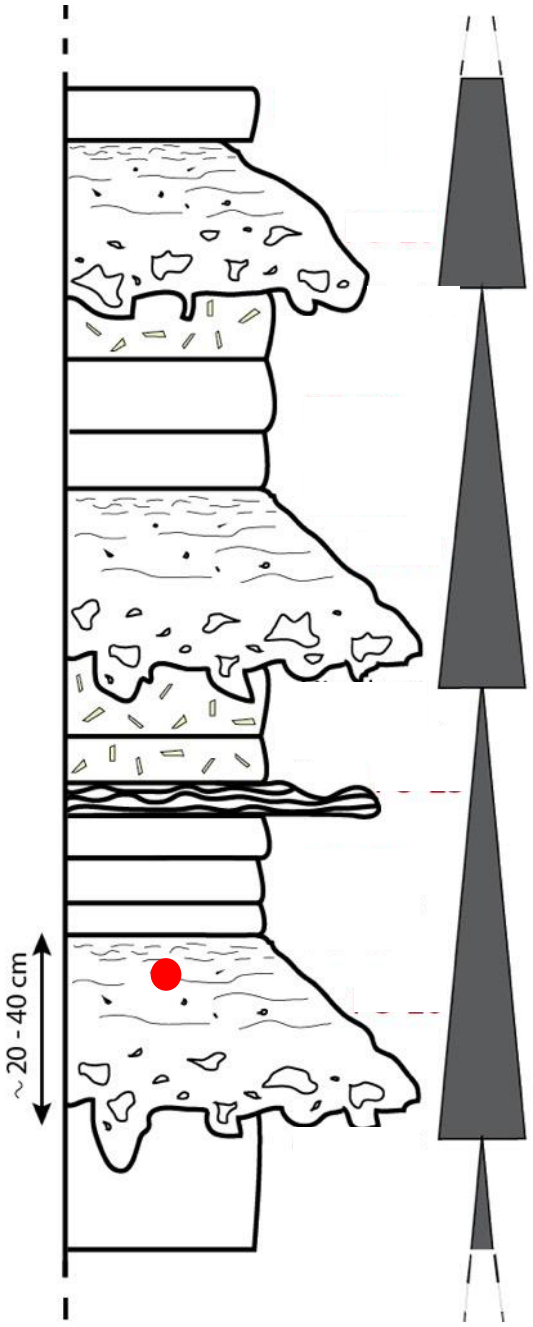




Note that the Upper Oxfordian is distinctly cyclic (left), with each cycle showing very similar vertical succession of facies, indicated by the numbers.

Complement your outcrop observations with the thin sections indicated on the left, as well as the example photos on the next few pages.

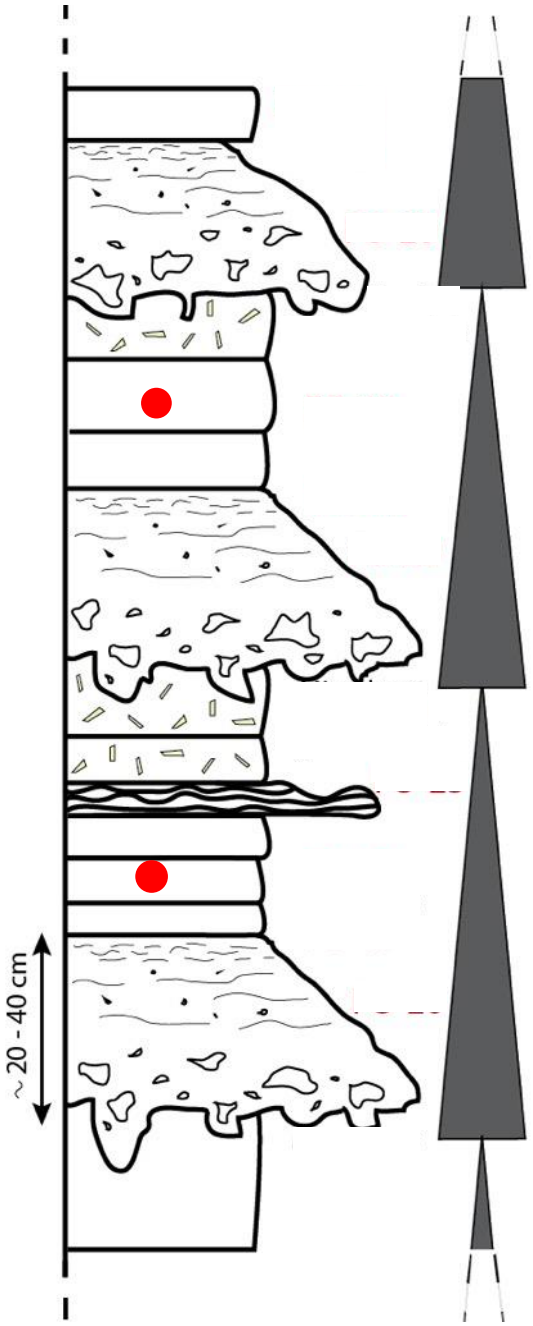
Take note of:
- Grains, their %, size and degree of preservation
- Any other minerals present
- Presence of sparite, micrite and/or porosity, and their %
- Texture
- What is the Dunham name for this thin section?





Sedimentary structures in the overlying sediment?
Implications for sedimentary processes?
Evidence for high-energy deposition overlying karst: shallow subtidal



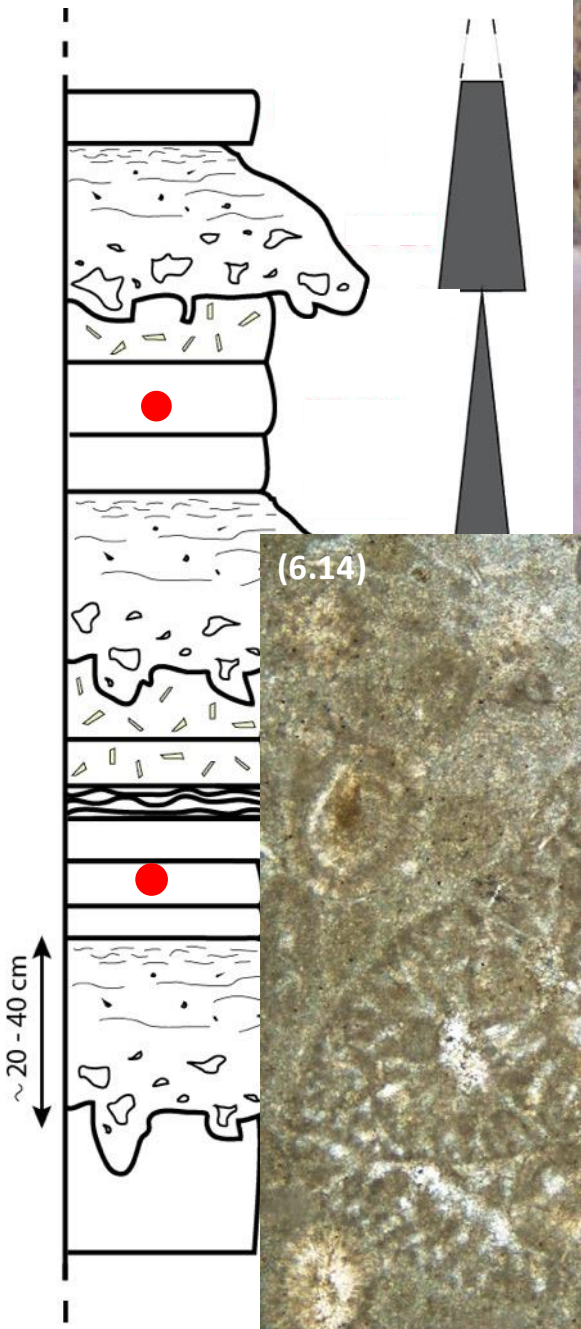




Identify this type of grain

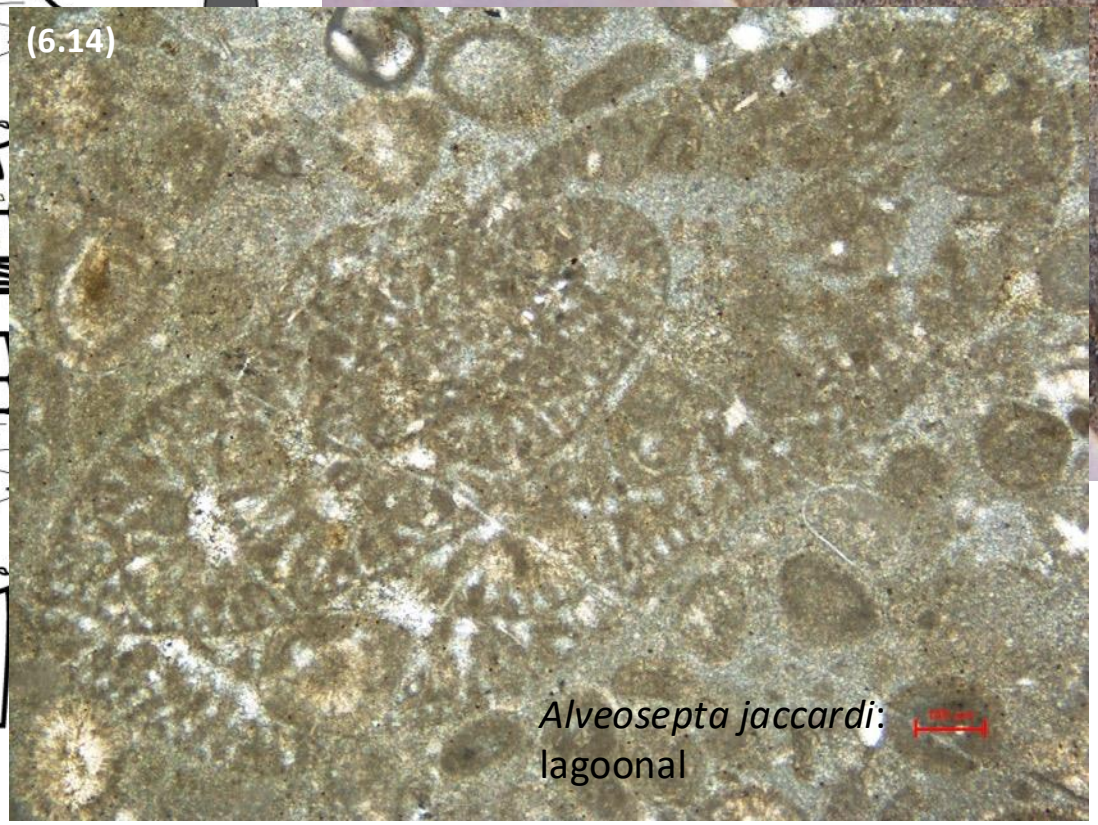
Oncoids – typical of lagoonal settings

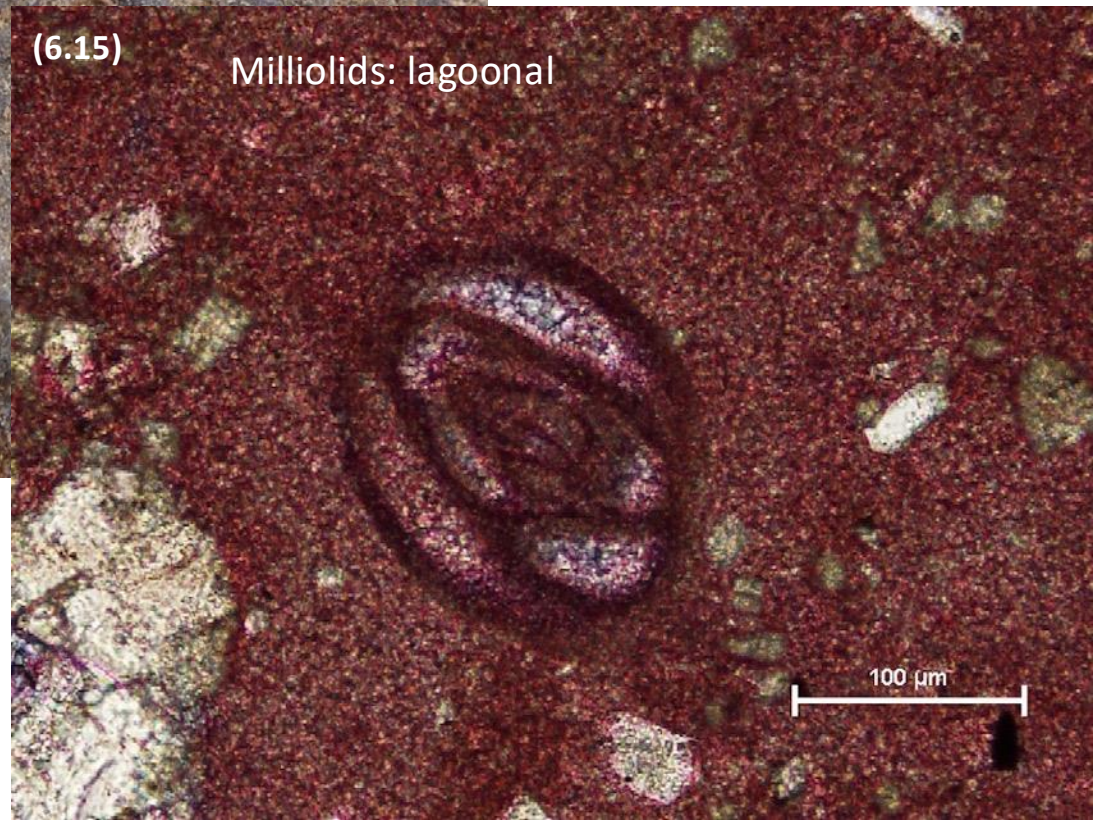
Grain types (TS) (1-2)

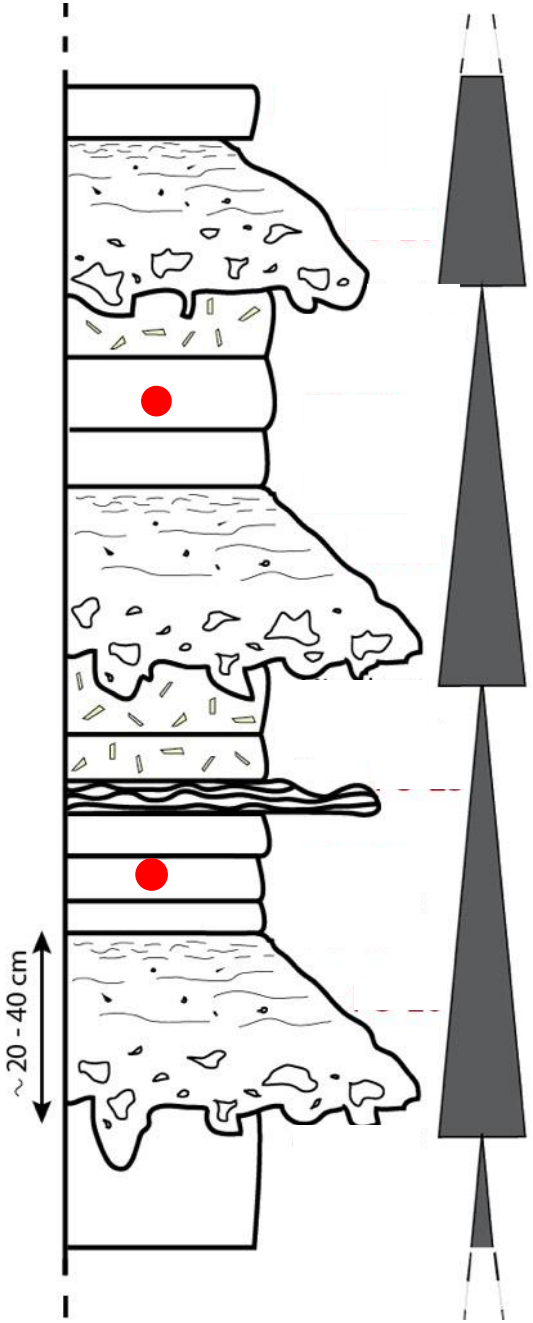


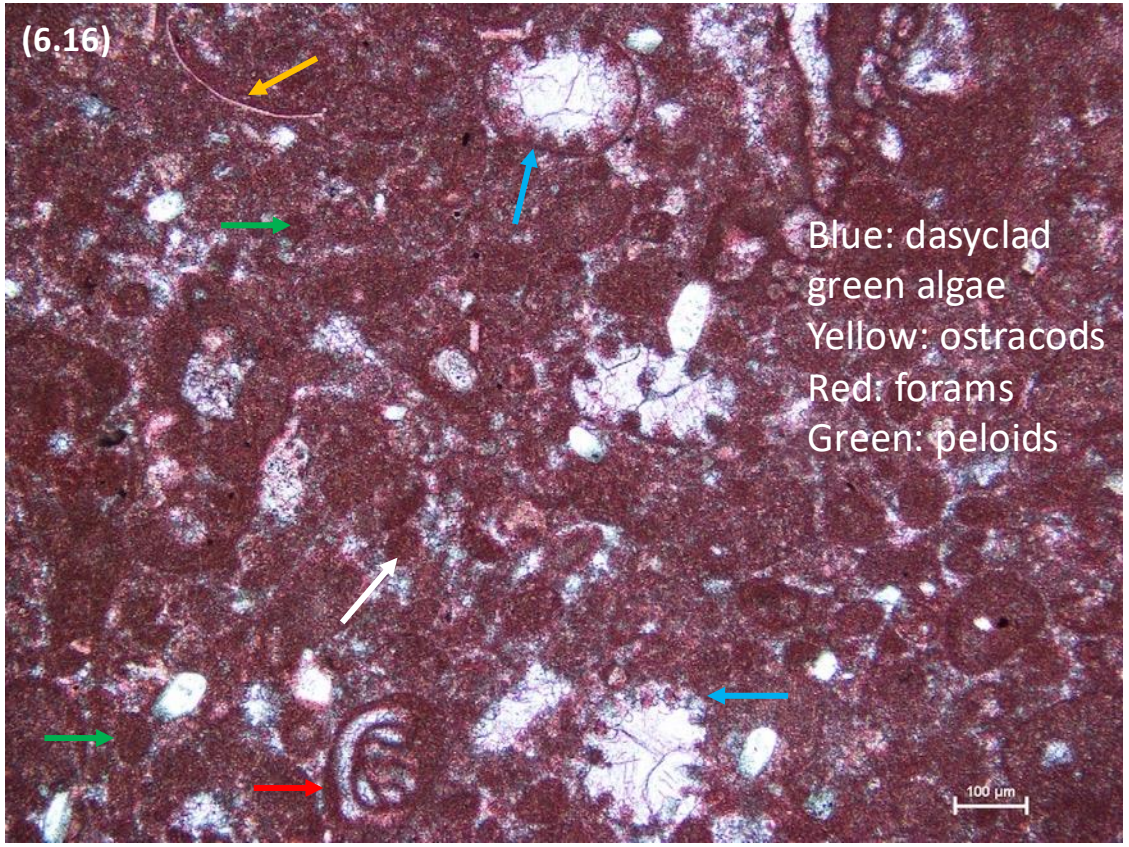


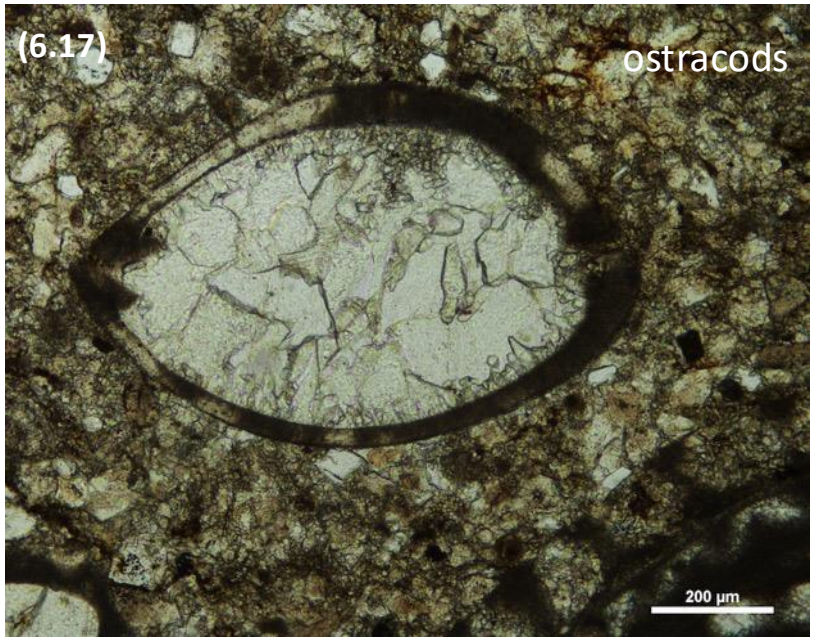
What are these grains?
foraminifera
What do they imply about the depositional environment?
Foraminifera typical of very shallow lagoonal environments











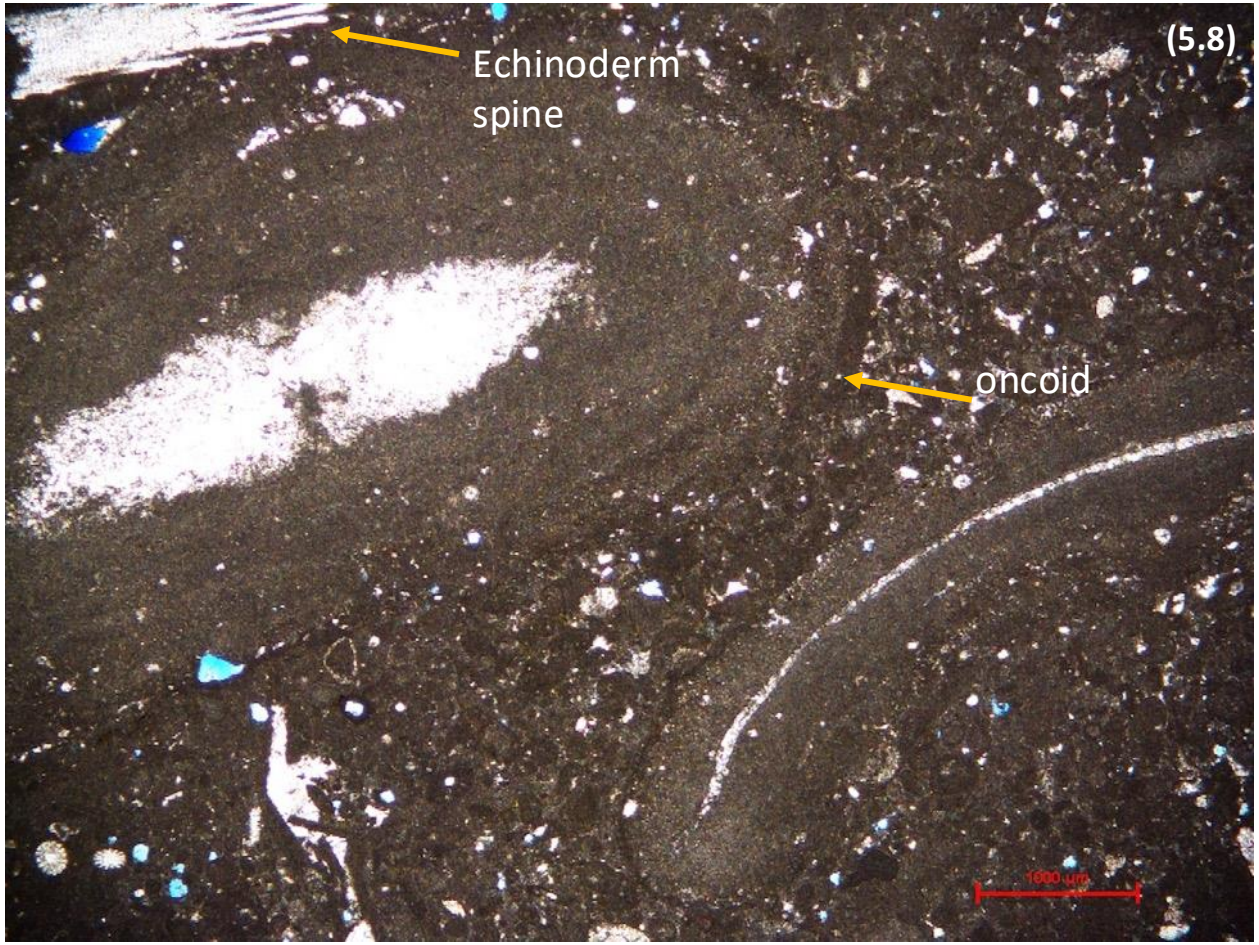


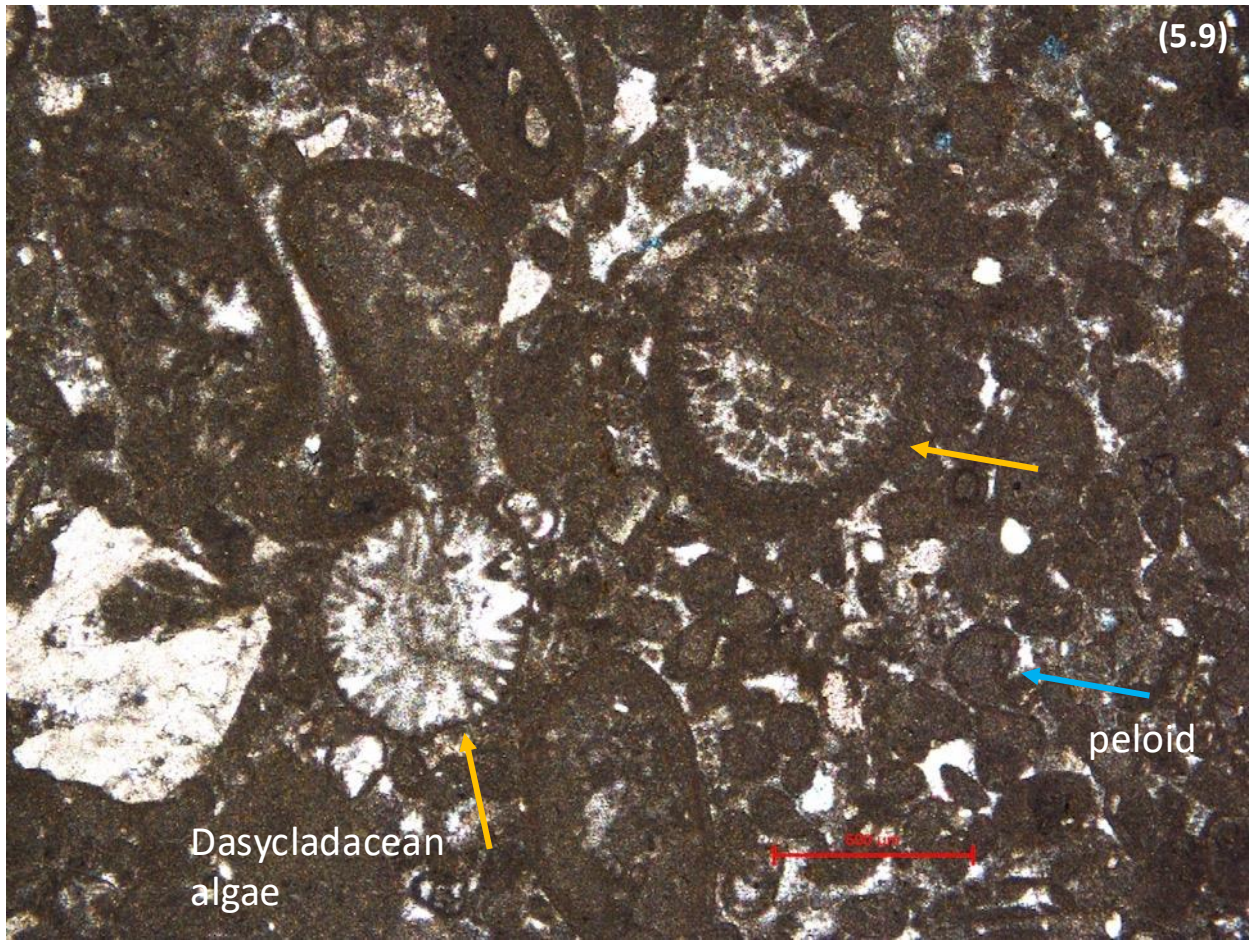
What are these grains?

Dasyclad algae, ostracods

What do they imply about the depositional environment?

Algae are typical of very shallow subtidal waters; ostracods are potentially widespread, but common in nearshore environments of variable salinity





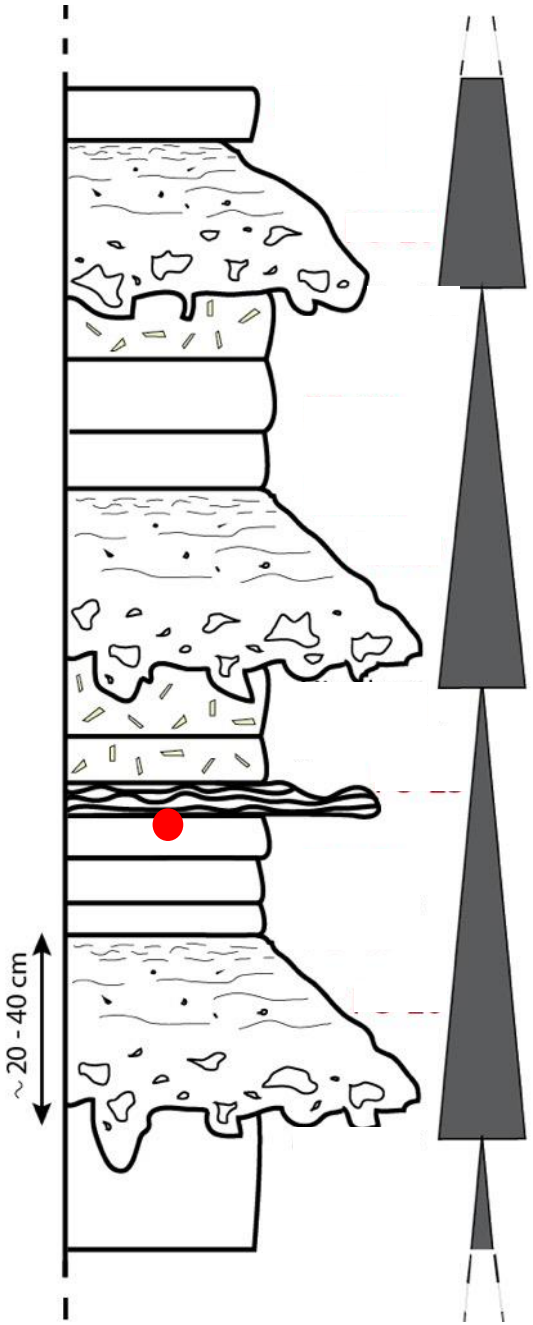
Identify these grains

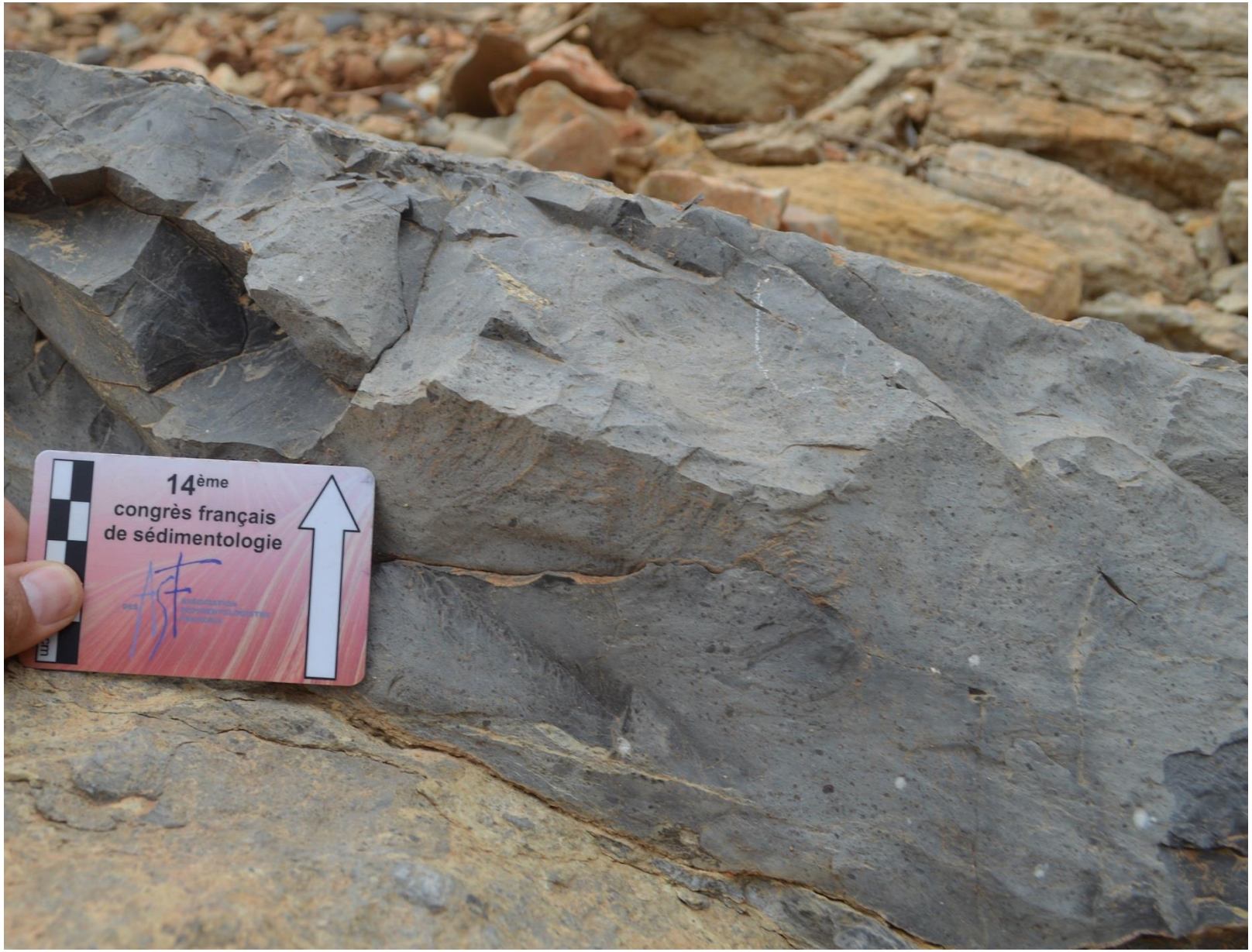
Oncoids, echinoderm fragments, peloids, dasycladacean (green) algae, packstone texture

What depositional processes and environment are indicated?

Dasyclads live in very shallow (<10-15m) water (euphotic zone); peloids and oncoids are typical of lagoons; echinoderms may be washed in (they prefer normal salinity) – note these are mostly fragments (cf Exercise 3)

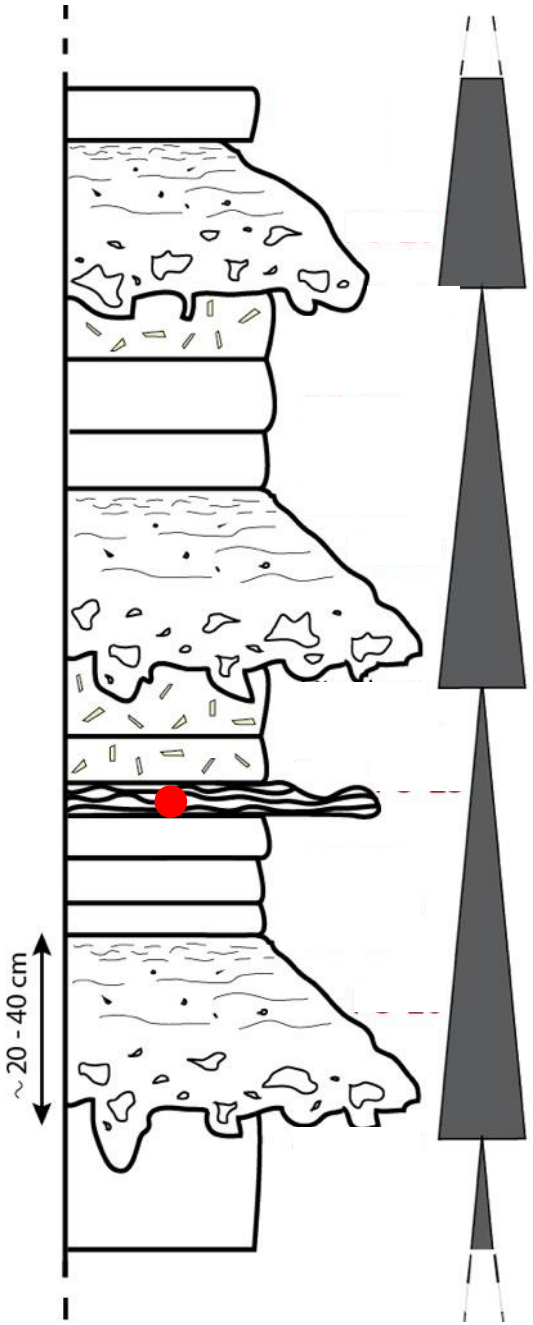
Outcrop appearance (HS) (2)

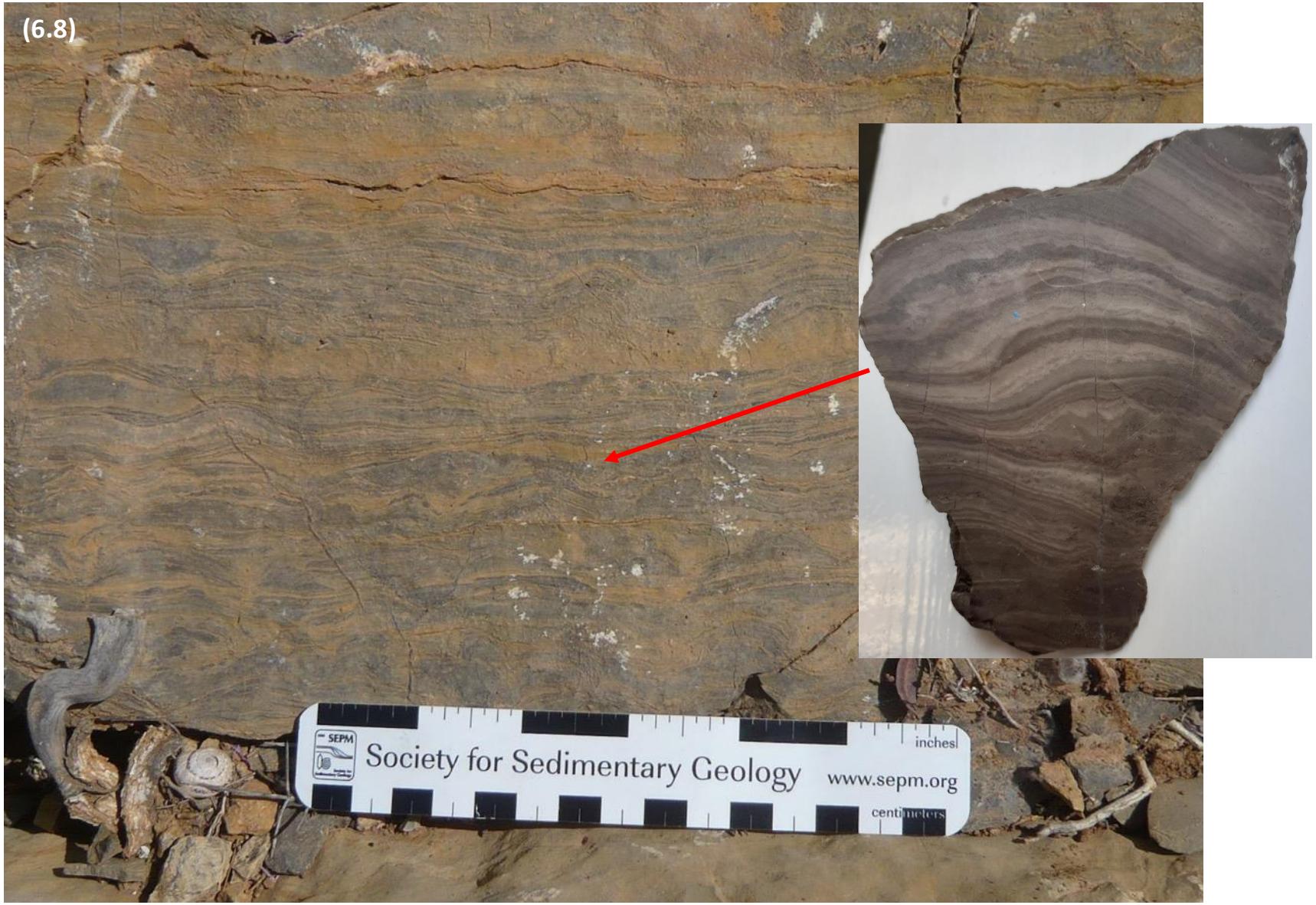




Textures in the middle part of the cycle? Implications for sedimentary processes?

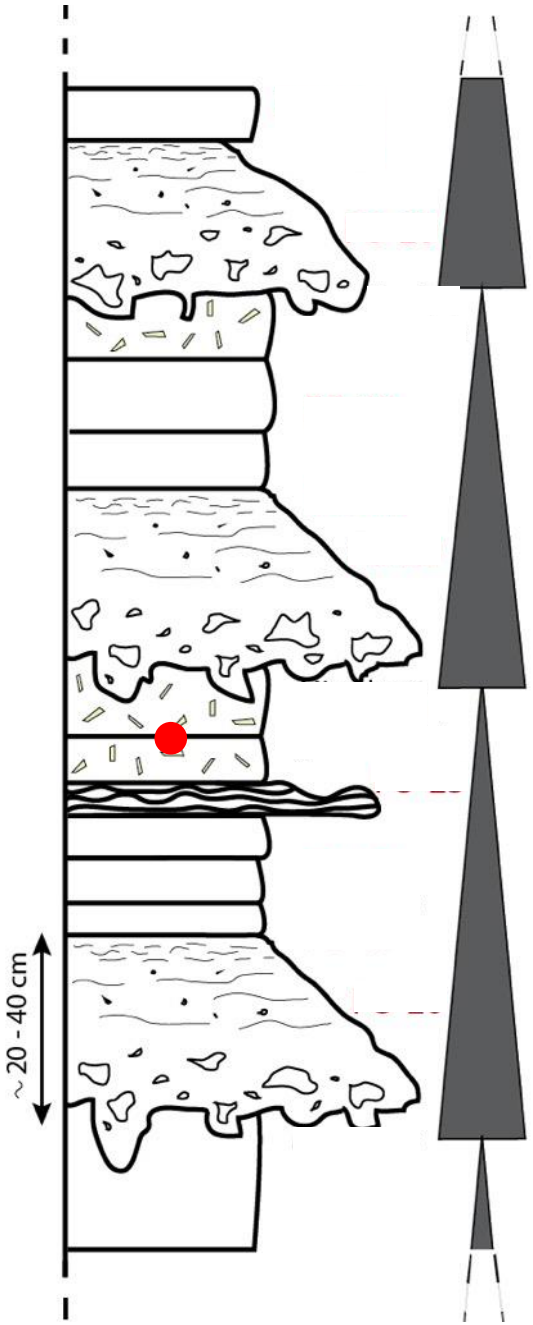
Mdst/Wkst – fine, so low energy deposition

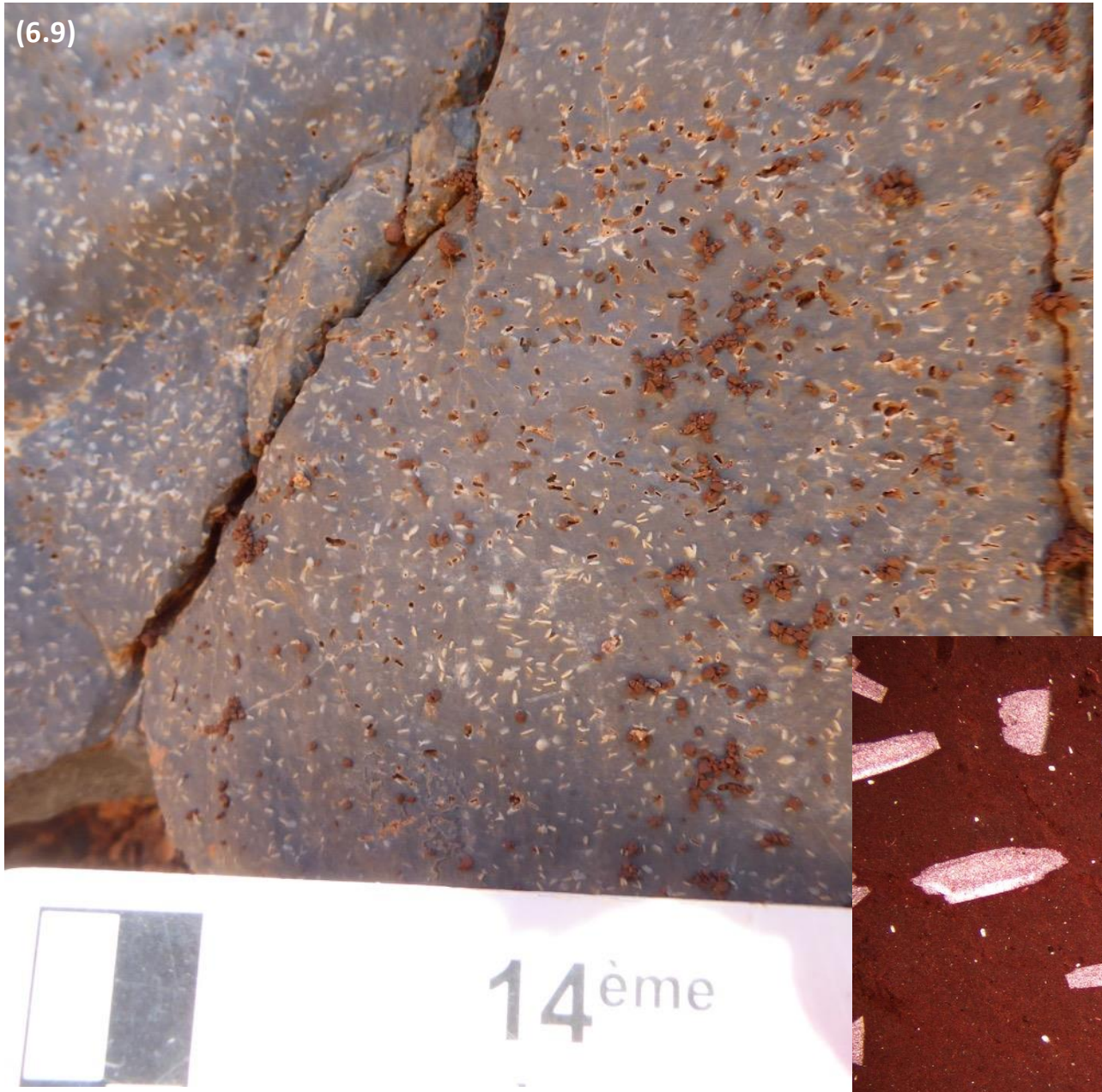




Sedimentary structures in the middle part of the cycle? Implications for sedimentary processes?

Stromatolites/microbial mats: intertidal



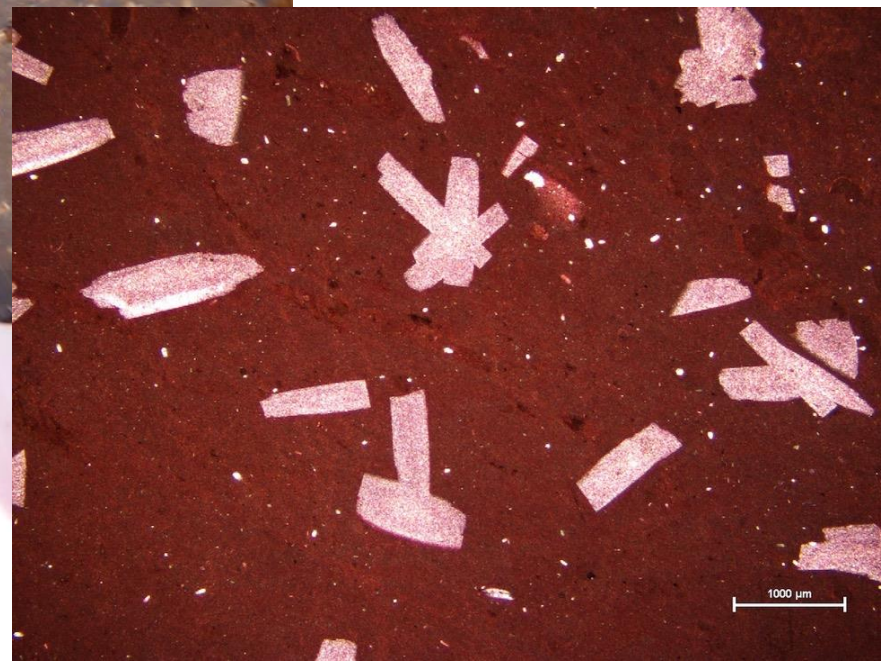


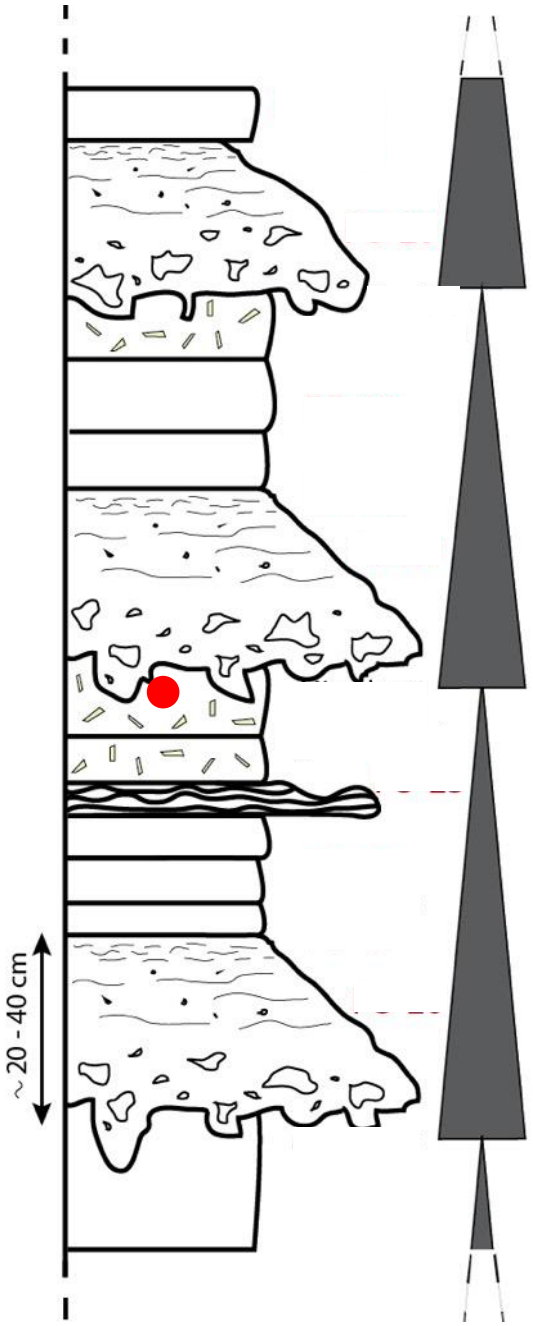
Outcrop appearance (HS) (4)

Sedimentary structures in the upper part of the cycle?
Calcite pseudomorphs after gypsum crystals

Implications for sedimentary processes?
Gypsum grew in micrite of lagoon/intertidal mud; evidence for episodically elevated salinities

Compare with thin section IMZ109:





Sedimentary structures in the upper part of the cycle?

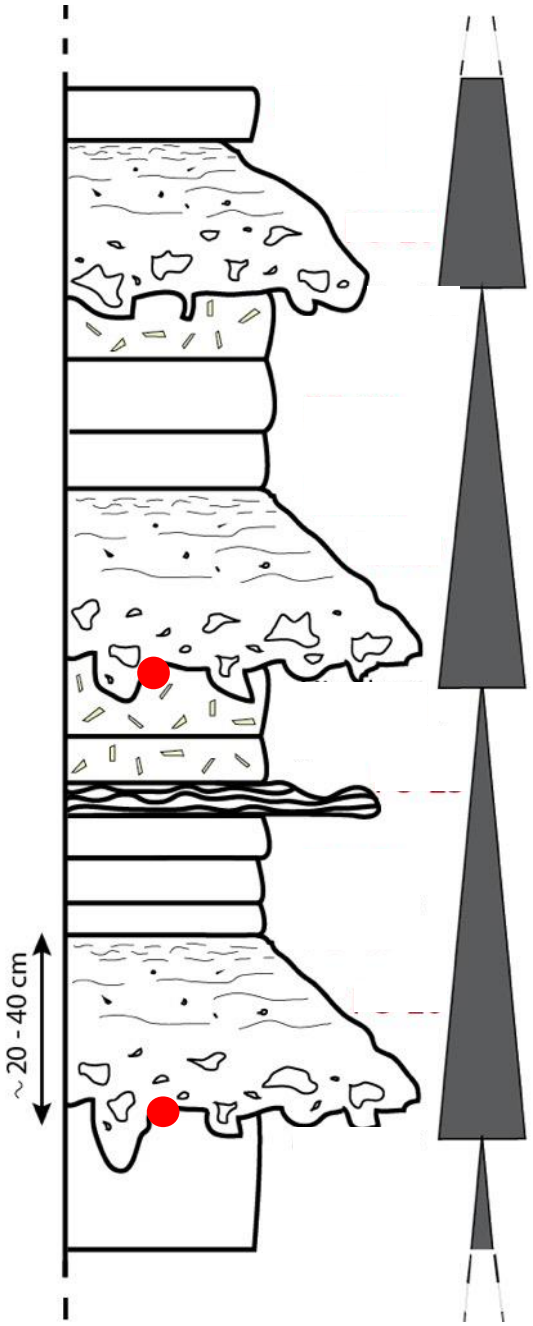
Mud cracks

Implications for sedimentary processes?

exposure



Outcrop appearance (HS) (5)

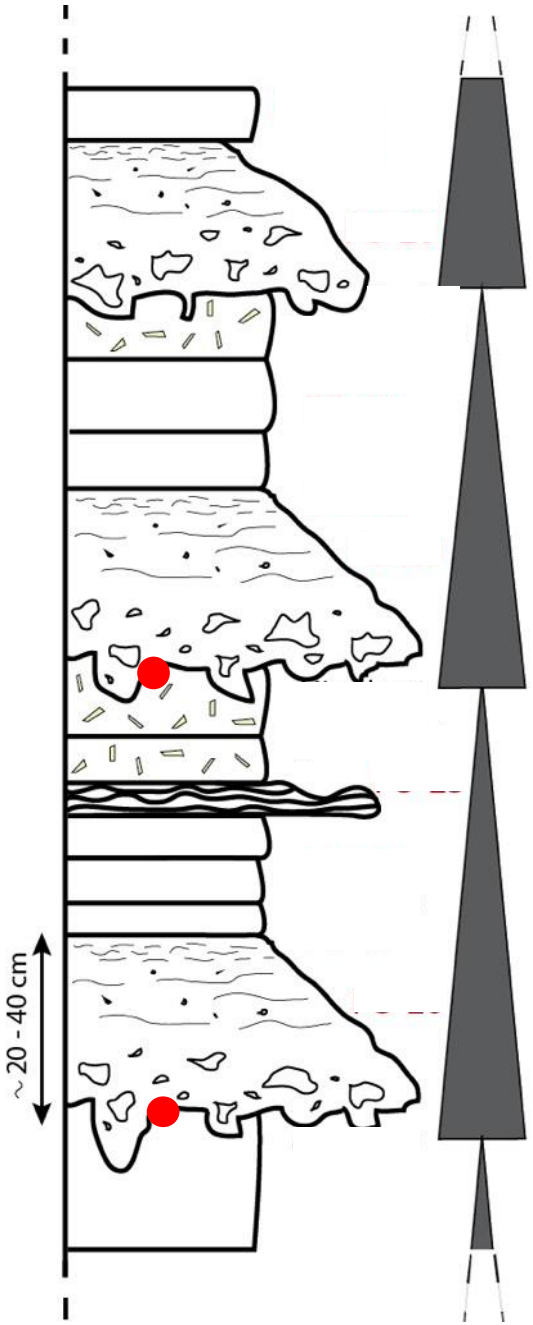




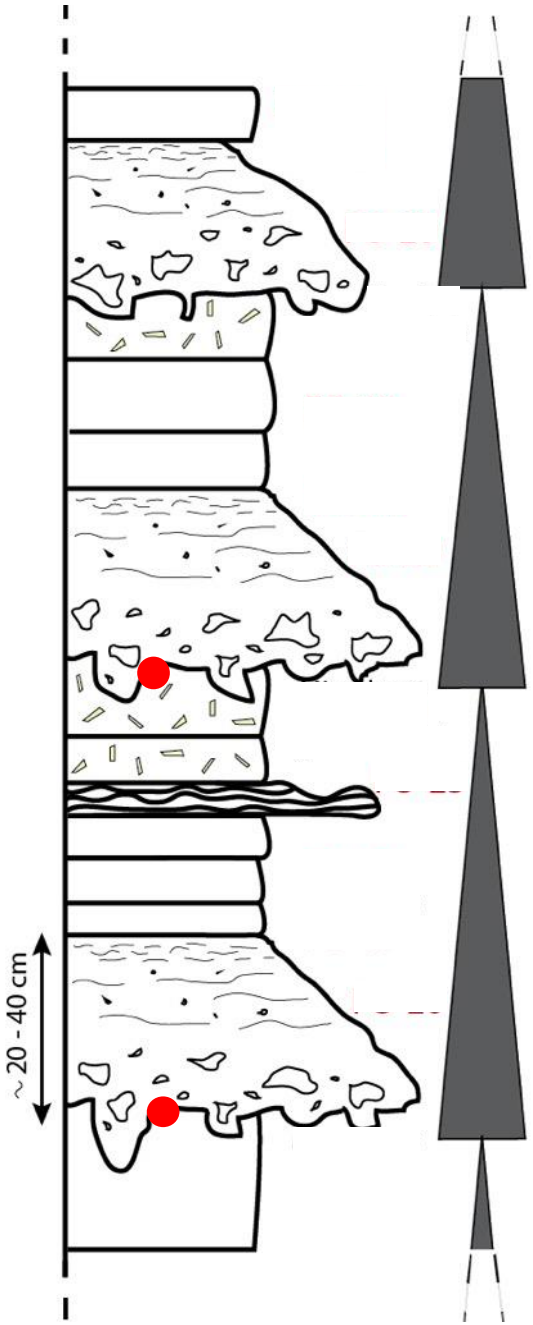
What is the nature of the basal surface?

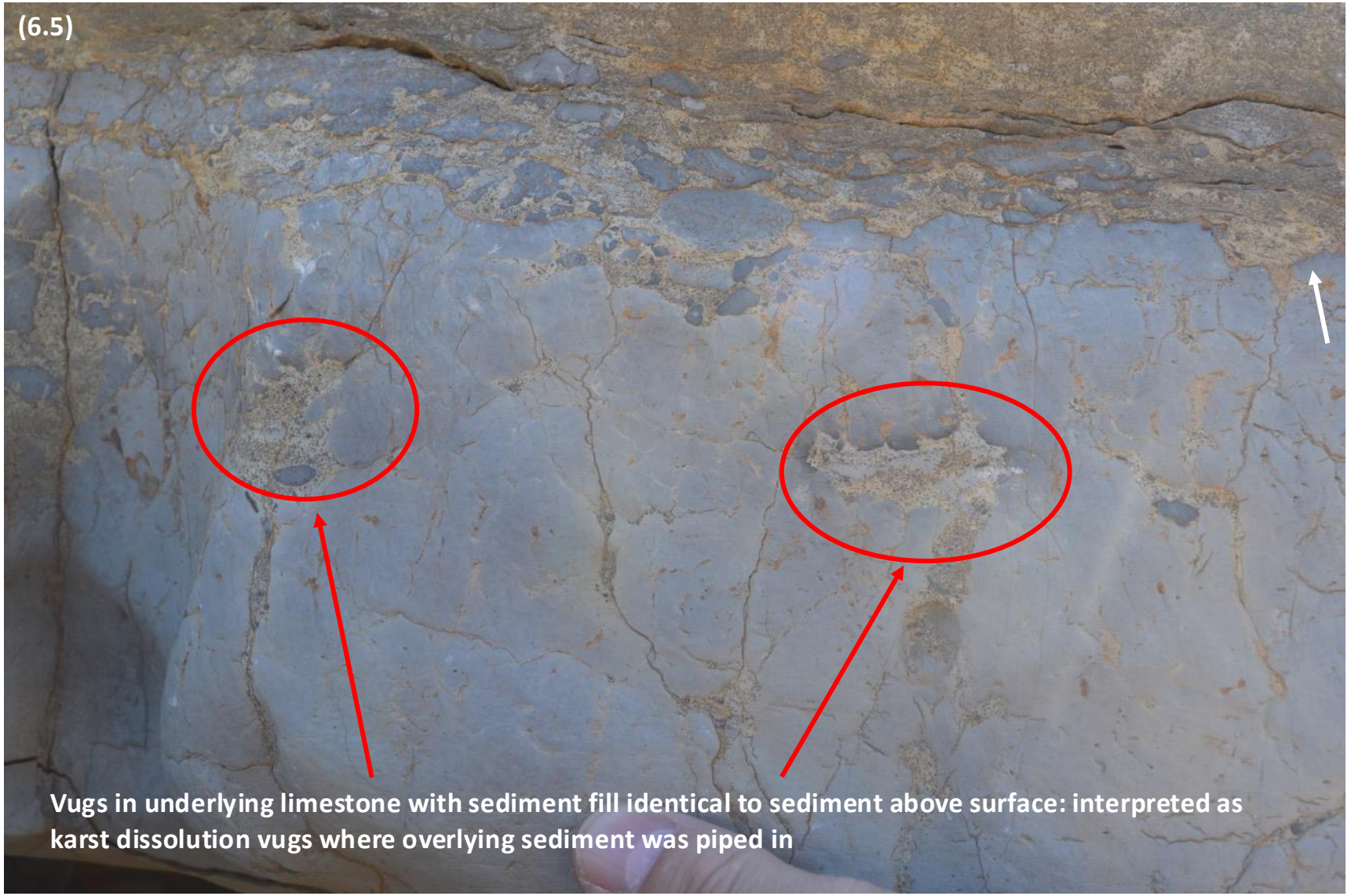


Outcrop appearance (HS) (5)



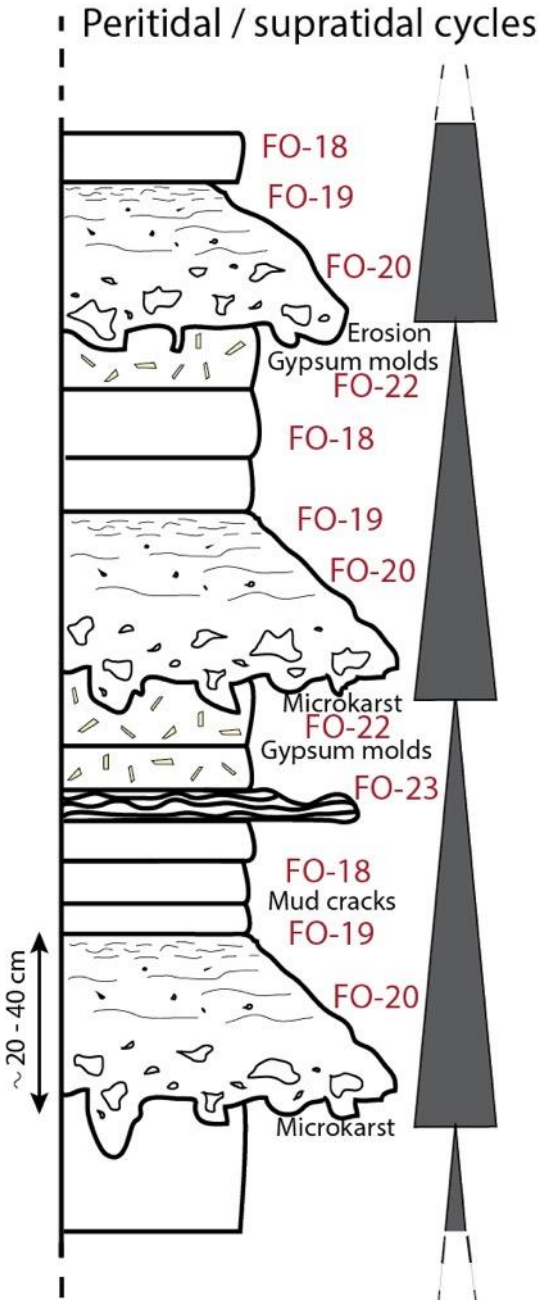


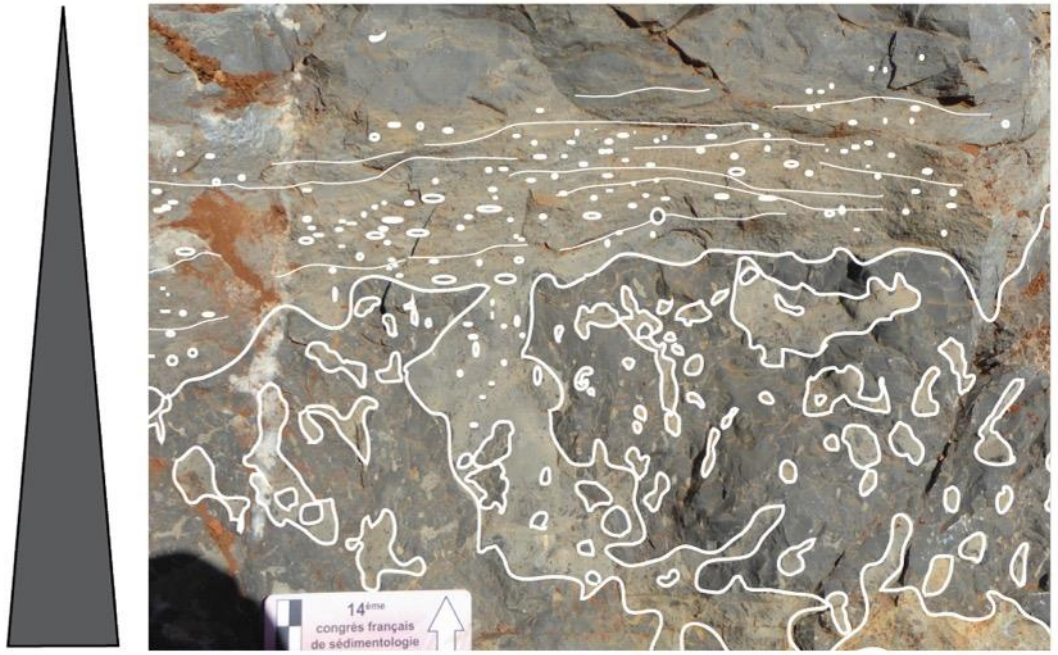


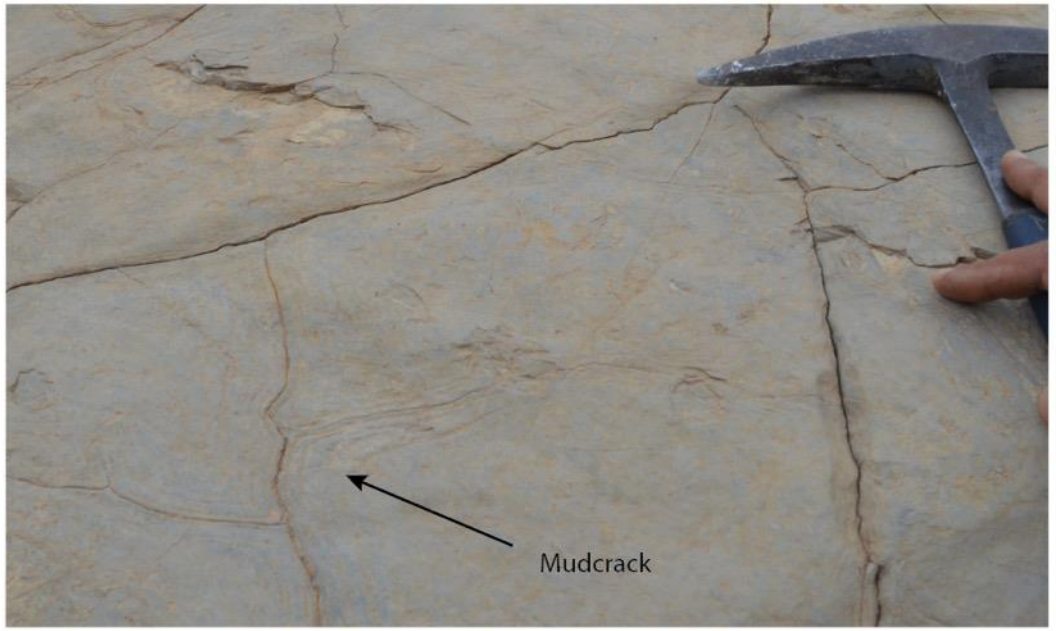


What is the nature of the basal surface?

Interpreted as subtle karst surface where underlying supra/intertidal mudstones were dissolved







Summarize your observations on this unit:

- What grains were identified?
- What sedimentary structures were identified?
- What is the texture of these beds?
- What sedimentary processes can you interpret?
- What environmental information is contained within the skeletal grains?
- Overall, what environment is likely?

Cycles consist of subtidal (very shallow lagoon) – intertidal (stromatolites, gypsum) – supratidal (karst, mudcracks) facies: typical of m-scale cycles of peritidal environments
Karst is quite subtle, so unlikely a fully humid climate